

<https://auriga-navi-1--michaelomosh118.replit.app>

Generating an APK locally is a great way to take full control of your mobile build process without relying on cloud credits or external servers. Since you are moving from Replit to a Local Build using Capacitor, here is a clear roadmap to get that APK in your hands.

1. Prepare Your Environment

Before you can run Android commands, your PC needs the right tools installed.

- * Node.js: To run npx commands.
- * Java Development Kit (JDK): Version 17 is currently recommended for modern Android builds.
- * Android Studio: This provides the SDK (Software Development Kit) and the build tools needed to compile the code.

2. The Migration Workflow

Once your tools are installed, follow these steps to move your code from Replit to a local Android project:

- * Download & Extract: Download your project from Replit as a .zip file and extract it to a folder on your PC.

- * Install Dependencies: Open your terminal (PowerShell or CMD) in that folder and run:
`npm install`

- * Build the Web Project: Capacitor needs a "dist" or "build" folder to wrap. Run your project's build command (usually `npm run build`).

- * Add Android Platform:
`npx cap add android`

- * Sync the Code: Every time you make a change to your web code, run:
`npx cap copy android`

3. Generating the APK in Android Studio

After running `npx cap open android`, Android Studio will launch your project.

- * Wait for Gradle: Look at the bottom of the window; wait for the "Gradle Sync" to finish. This can take a few minutes the first time.

- * Build the APK: 1. Go to the top menu: Build > Build Bundle(s) / APK(s).
2. Select Build APK(s).

- * Locate the file: Once finished, a notification will appear in the bottom right. Click "locate" to find your app-debug.apk.

Pro-Tips for Success

- * Environment Variables: Ensure your ANDROID_HOME path is set correctly in your system's Environment Variables so the terminal knows where your SDK lives.

- * Testing: Instead of just building the APK, you can plug in your Android phone via USB and click the "Run" (Green Play button) in Android Studio to see the app live on your device instantly.

Would you like me to walk you through how to set up those Windows Environment Variables for the Android SDK?